King Saud University
College of Computer and Information Sciences
Department of Information Technology

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Graduation Project Proposal

<Eyecept>

Prepared by

<Lana AlHumaidan, 444200238>
<Shahad Alotaibi, 444201168>
<Fatema AlHashem, 444203021>
<Joud Aljamaan, 444200201>

Supervised by
Dr.Ameera Almasoud

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Table of Contents

1 Introduction 3
2 The Problem 3
3 The Solution 4
4 Product Vision 4
5 Product Roadmap 6
6 Objectives 6
7 Scope 9
8 Hardware/Software Tools and Cost 10
9 Scrum Team 11
9.1 Skill Set Requirements 11
9.2 Roles and Responsibilities 11
10 References 12

1 Introduction

In high-value environments such as museums and jewelry exhibitions, security is not only about monitoring spaces, but also about protecting valuable assets, cultural heritage, and ensuring visitor safety. Because valuable items are often clearly visible and easily accessible, these places naturally become attractive targets for theft attempts. Although modern surveillance cameras are widely used, most existing systems still focus mainly on recording events after they happen, rather than understanding human behavior or supporting early intervention.

In reality, theft rarely happens suddenly. It is usually preceded by small but meaningful behaviors, such as staring at a specific object for a long time, repeatedly turning the head toward it, or slowly and gradually moving the hand closer to a restricted area. These signals may appear within seconds or minutes, yet they are often missed, especially in crowded environments where security personnel must monitor multiple screens at the same time. This pressure increases the cognitive load on guards and creates a real gap between observation and early preventive action.

In addition, many current intelligent surveillance systems rely on cloud-based processing, which raises concerns related to privacy, data security, and response time, particularly in sensitive environments. This has created a growing need for intelligent solutions that operate locally, analyze data in real time, and ensure that visual information never leaves the facility.

To respond to these challenges, this project proposes Eyecept, an intelligent edge-based visual surveillance system designed to detect early behavioral indicators of suspicious activity before a theft attempt occurs. By analyzing head orientation, visual attention duration, and gradual hand movement, Eyecept aims to transform surveillance systems from passive recording tools into proactive security solutions that support early decision-making. This document presents the problem that motivated the development of Eyecept, the proposed solution, the project objectives, scope, and overall system structure.

2 The Problem

In museums and jewelry exhibitions, surveillance systems are mainly used to observe visitor movement and record events. While these systems are effective in documenting obvious actions and assisting guards in monitoring, they still lack the ability to understand subtle behaviors that may indicate suspicious intent before a theft occurs.

Real-world observations show that many theft cases begin with minor and easily overlooked signs, such as repeatedly turning the head toward a valuable item, maintaining visual attention for an unusual duration, or gradually moving the hand closer to restricted zones before any physical contact happens. Although some academic and commercial solutions have explored motion tracking and body pose estimation, there remains a clear gap in their ability to interpret head orientation as focused attention, measure attention duration, and distinguish between natural curiosity and potentially suspicious behavior.

Moreover, a large number of existing intelligent surveillance systems depend on continuous internet connectivity and cloud infrastructures, which introduces risks related to privacy exposure, delayed responses, and system security. In high-value and privacy-sensitive environments, there is an increasing demand for intelligent systems that operate locally and provide real-time analysis without sending data outside the facility.

Therefore, the core problem addressed in this project is the lack of intelligent systems capable of understanding subtle pre-theft behavioral signals and enabling early preventive intervention while preserving privacy. Eyecept is designed to fill this gap by providing an edge-based surveillance system that moves beyond simple motion detection toward deeper behavioral understanding.

3 The Solution

In high-value environments such as museums and jewelry exhibitions, most surveillance systems focus on recording incidents after they happen, which limits the ability of security teams to respond early. Subtle behaviors that often appear before a theft attempt are usually overlooked, especially in crowded areas.

For example, a visitor may stand in front of a valuable item, keep their head and gaze directed toward it for an extended period, and slowly move their hand closer to the display. Since no clear violation has occurred, traditional systems do not flag this behavior, even though it may indicate a potential theft attempt.

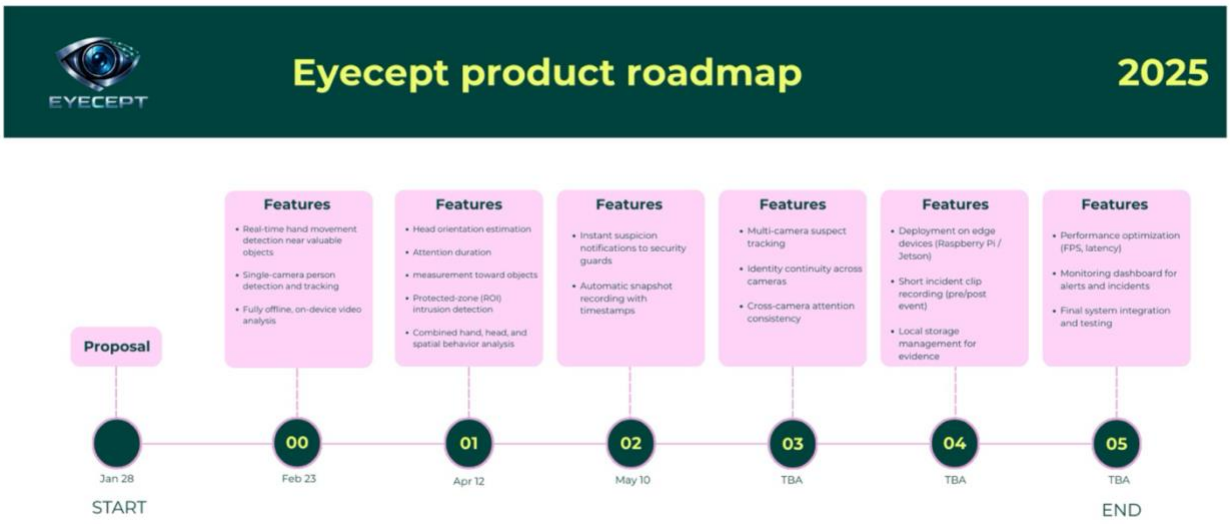
The Eyecept Guardian system addresses this issue by focusing on a small set of reliable pre-theft indicators: hand movement, head orientation, and attention duration. These behaviors are analyzed locally using computer vision techniques, without relying on cloud processing. When unusual patterns are detected, the system sends a real-time alert to security staff, enabling early intervention while maintaining visitor privacy.

4 Product Vision

The product vision is "*a brief statement of the desired future state that would be achieved by developing and deploying a product. A good vision should be simple to state and provide a coherent direction to the people who are asked to realize it*." [1]

Product Vision:
For a museum and jewelry exhibition security teams
Who require early detection of suspicious behavior
The Eyecept is a web-based intelligent surveillance system
That detects pre-theft behaviors in real time
Unlike traditional CCTV systems like Avigilon Video Analytics
Our product analyzes subtle behaviors locally while preserving privacy

5 Product Roadmap



6 Objectives

Product (customer focus-value): SmartCam Guardian is designed to overcome one of the most significant weaknesses of a conventional surveillance system: the inability to identify early pre-theft activity and prevent a real robbery. Current camera systems are primarily passive surveillance devices capable of recording events once they occur, with little preventive benefit.

This product will be designed to offer proactive security by detecting subtle behavioral indicators, including focused attention, hesitation, and slow approach, which are usually the precursor of theft attempts in high-value settings. Moreover, SmartCam Guardian allows all video analysis to be conducted both on-device and offline, which allows quick response and removes the use of cloud infrastructure.

Security guards, museum and exhibition administrators, and jewelry store owners are the main beneficiaries of SmartCam Guardian as they can have earlier knowledge of suspicious activity and intervene before it is too late. Another indirect benefit of the visitors is increased safety without privacy loss, since video data is not sent to external servers and instead is stored in the facility in its entirety.

The product objectives of SmartCam Guardian are as follows:

1. **Enable real-time hand movement detection**
   Identify and detect hand movements around valuable items to detect suspicious activities before the physical contact.

2. **Analyze head orientation and attention duration**
   Track head pose and the length of time an individual focuses on a specific item to infer sustained interest that may indicate pre-theft intent
3. **Detect protected-zone intrusions**
   Establish sensitive spatial areas around exhibits of high value and raise alerts in case hands or bodies intrude into these spaces too soon.
4. **Provide instant suspicion notifications to security guards**
   Send real-time notifications in case suspicious behavior patterns are identified to facilitate more efficient and quick human reaction.
5. **Ensure on-device, offline video processing**
   Perform all analysis locally on edge devices to preserve privacy, reduce latency, and eliminate dependence on cloud connectivity.
6. **Support multi-camera suspect tracking**
   Keep constant surveillance of a target of interest on several camera perspectives within the facility.
7. **Documented proof in the form of automatic snapshot or short incident video.**
   Record and hold timestamped image evidence on to the local machine to be reviewed, reported, and analyzed after the fact.

- **Project (solution focus-plan):**

The SmartCam Guardian project will require the following steps to be complete:

1- **Requirements analysis and use-case definition:** Find out the security requirements in museums and jewels exhibition, pre-theft behavior conditions, secure areas, and alarm states.
2- **Selection and preprocessing of online datasets:** It concerns the selection of appropriate publicly available surveillance databases, screening of the pertinent situations, and an initial processing of video data (frame extraction, labeling refinement, and quality correction) to fit the specific environment.
3- **Model selection and integration:** Pre-trained models of hand detection, head pose and person tracking need to be integrated and be trained on CCTV-like video.
4- **Development of behavior analysis logic:** Develop logic to analyze hand movement, head orientation and attention duration and use this information to determine early suspicious behavior.
5- **Implementation of zone and intrusion detection:** Design value-based areas of interest (ROIs) around valuable objects and identify the intrusion prior to a physical contact.
6- **Evidence and alerting system:** Introduce real-time notifications to the security guards and automatically record snapshots or short clips with timestamps.
7- **Edge deployment and performance optimization:** Deploy the system on an edge device and optimize performance for offline, on-device processing.

8- **Monitoring dashboard development:** Develop a simple web-based dashboard to visualize alerts, events, and recorded incidents.
9- **Testing, evaluation and refinement:** Test the system with a variety of cases and camera views, determine the accuracy and latency, and make changes to the system.
10- **Documentation and final integration:** Final preparation and documentation of the system and prepare the system to final demonstration and evaluation.

- **Learning (student focus):**
  1- **Computer vision on surveillance systems**
  Practical expertise in integrating and deploying statistical models of computer vision like YOLO to detect objects and hands, Hopenet to estimate the head pose, and ByteTrack to track people in actual CCTV applications.
  2- **AI system design based on behavior**
  Developing the ability to transform raw model outputs into useful behavioral attributes through integration of a series of indicators (hand movement, attention duration and spatial context) to identify the early signs of suspicious behaviour.
  3- **Edge AI and on-device processing**
  Practical experience deploying AI models on edge devices, optimizing performance for real-time, offline processing, and understanding the trade-offs between cloud-based and local computation.
  4- **Multi camera coordination and tracking**
  The knowledge of how several cameras can team up and follow a target subject in various positions and still have the same identity throughout an area under surveillance.
  5- **Alerts and notification in real-time**
  How to create and establish event-driven alerts and push notifications to alert the security staff immediately in case suspicious activity has been detected.
  6- **Local event logging and storage of data**
  Capability to achieve experience with the local storage of snapshots, incident clips, and alert metadata, as well as data flow between the processing of AI, storage, and monitoring dashboard.
  7- **Agile system integration and planning**
  Training to break down a complex system into manageable modules, scheduling work into sprints, managing progress on a task management tool and incorporation of multiple components into a working system.

7 Scope

The Eyecept project aims to enhance security in museums and jewelry exhibitions by detecting early suspicious behavior using video analysis. The system will process video locally from USB or IP cameras and provide real-time alerts to security personnel. It will initially support the English language and operate as an academic prototype in controlled environments. The project focuses on behavioral detection (head orientation, hand movement, and protected-zone intrusion) and does not include identity recognition or large-scale commercial deployment. The scope may be refined as development progresses.

8 Hardware/Software Tools and Cost

All hardware and software tools required to develop the solution should be specified in this section by completing the table below. For software tools, please include IDE tools you will be using and any other APIs required. (*do not include day-to-day communication tools such as WhatsApp, etc.*)

Hardware Tools	
Name and Description	Cost *mention date it will be available*
Laptop for coding	No cost
USP or IP camera	250–500 SAR
Raspberry Pi 5 / Jetson Nano for edge processing	800–1500 SAR
External SSD storage	400-700 SAR
Stable network connection for alerts	200-600 SAR
Software Tools	
Name and Description	Cost
Word for documentation	No cost
Jupiter Notebook or google Collab	No cost
Python 3.10+	No cost
PyTorch	No cost
OpenCv	No cost
Ultralytics YOLO	No cost
ByteTrack or Norfair (tracking)	No cost
Hopenet (head pose estimation)	No cost
NumPy / Pandas	No cost

Please mention other costs (if any) that are relevant to the project

9 Scrum Team

9.1 Skill Set Requirements

List the set of technical skills required for the team to be able to develop the product, also highlight any skill gaps available and describe how will the team address this need.

Technical Skill Required	What is the current level of the team (*beginner-intermediate- advanced*) for each skill? How will the gap be bridged? *(if necessary*) Learning plan
Python programming	Intermediate
Computer vision	Intermediate
Time management	Advanced
Problem solving	Advanced
Teamwork & communication	Advanced

9.2 Roles and Responsibilities

Scrum Team	
Product Owner: (list name of the project supervisor)	Dr. Ameera Almasoud
Developers: (list student names)	Fatema AlHashem Joud Aljamaan Shahad Alotaibi Lana AlHumaidan
Scrum Master (SM): (list name of designated SM for the project)	Dr. Ameera Almasoud
Stakeholders: (list names of project stakeholders, include customer names if any)	Dr. Ameera Almasoud, Examination team

10 References

List any references you mention in your document here

[1] Rubin, K. S. (2017). *Essential Scrum: A practical guide to the most popular agile process*. Upper Saddle River, NJ: Addison-Wesley.